

A PROJECT REPORT ON

**Smart Agriculture System Using Machine
Learning, Blockchain**

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ABSTRACT

Agriculture is a vital sector that sustains economies and societies worldwide, yet farmers face numerous challenges in making informed decisions to maximize crop yield and quality. This project aims to develop a comprehensive web application that empowers farmers with the tools and information needed to make data-driven agricultural decisions. The application will provide several critical features: Crop Disease Detection: Utilizing OpenCV, the app will allow farmers to upload images of their crops for automated diagnosis, identifying potential diseases for early intervention. Soil-Based Crop Recommendation: The app will analyze soil nutrients, such as nitrogen and phosphorus, and recommend optimal crops that are best suited for the soil composition, improving crop yield and productivity. Weather Forecasting: By integrating the Google Weather API, the app will deliver real-time weather forecasts, enabling farmers to plan their agricultural activities effectively and reduce crop damage due to unexpected weather conditions. Government Schemes Information: The app will provide detailed information on government policies, schemes, and subsidies available to farmers, ensuring they have access to the resources they need. Blockchain Integration : Smart contracts were developed in Solidity for crop transactions and traceability. These contracts are deployed on the Ethereum blockchain and accessed using a Web3.js-based interface. Transaction details are stored immutably, and integration with IPFS ensures decentralized document storage. This multifaceted approach combines modern technologies such as computer vision, machine learning, blockchain, and real-time data integration to offer a practical solution for farmers. By implementing these features, the application will enhance agricultural productivity, reduce crop loss, and ensure farmers have access to crucial information and resources.

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CHAPTER 1

INTRODUCTION

1.1 Overview

The Smart Agriculture System is a project designed to address critical challenges faced by the agricultural sector using advanced technologies such as machine learning, blockchain, and IoT (Internet of Things). With the global agricultural landscape under pressure from factors like population growth, climate change, and resource scarcity, the need for efficient and sustainable farming practices has never been greater. This system aims to empower farmers by providing tools to:

- Predict Optimal Crops:** Using machine learning models, the app analyzes soil nutrient levels (Nitrogen, Phosphorus, Potassium) and environmental data (temperature, humidity, rainfall) to recommend the most suitable crops for specific soil types, enhancing productivity.
- Detect Crop Diseases:** Farmers can upload images of their crops, and the app processes them using OpenCV-based image segmentation techniques to identify diseases early, enabling timely intervention and reducing crop loss.
- Provide Real-time Weather Updates:** By integrating with the Google Weather API, the app delivers real-time weather forecasts, helping farmers plan agricultural activities such as irrigation and harvesting, thus minimizing the impact of adverse weather conditions.
- Decentralized Crop Transaction Platform:** We developed a blockchain-based platform to facilitate secure, transparent transactions between farmers and buyers. Smart contracts written in Solidity handle automatic deal execution on the Ethereum test network. Web3.js enables frontend interaction with blockchain, while event listeners track transaction status in real time, improving traceability and trust.
- Access to Government Schemes:** The app provides up-to-date information on various government policies, schemes, and subsidies, allowing farmers to avail themselves of the necessary financial support and resources for improving their farming operations.

In essence, the Smart Agriculture System aims to modernize traditional farming practices through data-driven decision-making, increased transparency in the agricultural supply chain, and enhanced resource management. The system can significantly improve crop yields, reduce losses due to diseases, and ensure that farmers have access to the right tools and information at the right time.

1.2 Motivation

Agriculture plays a crucial role in sustaining economies and societies worldwide, yet farmers continue to face numerous challenges in making informed decisions that maximize crop yield and quality. To address these issues, this project proposes a comprehensive web application that empowers farmers

with modern technological tools and real-time information to make data-driven agricultural decisions. By leveraging cutting-edge technologies like machine learning, computer vision, blockchain, and real-time data integration, the app aims to enhance agricultural productivity, minimize crop loss, and ensure access to critical resources and information

1.3 Problem Definition and Objectives

1. Crop Disease Detection: Using OpenCV, farmers can upload images of their crops for automated diagnosis, allowing early detection of potential diseases. This feature ensures timely interventions, minimizing crop damage and improving yield. 2. Soil-Based Crop Recommendation: The app will analyze soil nutrients like nitrogen and phosphorus and recommend the most suitable crops based on soil composition, thereby optimizing productivity and sustainability. 3. Weather Forecasting: Integrating the Google Weather API, the application will provide real-time weather forecasts. This feature will enable farmers to plan their agricultural activities better and protect crops from adverse weather conditions. 4. Government Schemes Information: Farmers will have access to detailed information on relevant government policies, schemes, and subsidies. This ensures they can take full advantage of available resources to support their agricultural activities. 5. Traditional agricultural transactions lack transparency and are prone to exploitation by intermediaries. Our objective is to implement a decentralized platform using blockchain that enables secure, traceable, and trustless crop transactions between farmers and buyers.

1.4 Project Scope & Limitations

The Smart Agriculture System seeks to modernize farming practices through the application of cutting-edge technologies such as machine learning, image processing, blockchain, and IoT. The key focus areas of the project include: Crop Prediction: Leveraging machine learning models to analyze soil data (NPK levels, pH) and environmental conditions (temperature, humidity, rainfall) to recommend optimal crops for specific fields. This improves productivity and promotes sustainable farming practices. Crop Disease Detection: Utilizing OpenCV and image segmentation to allow farmers to upload images of crops for automated disease detection. This early identification can prevent large-scale crop losses by enabling prompt intervention. Weather Forecasting: Integrating the Google Weather API to provide farmers with

real-time weather data, enabling them to plan their activities (e.g., planting, irrigation, harvesting) based on current and predicted weather conditions. **Access to Government Schemes:** Providing detailed information on government agricultural schemes, subsidies, and policies, ensuring that farmers are aware of and can access available financial support and resources. **Notifications and Reminders:** The app will notify farmers about upcoming weather changes, disease risks, or government schemes to assist in timely decision-making.

Limitations

Internet Dependency: The system relies heavily on internet connectivity for real-time weather forecasting, blockchain verification, and data uploads (e.g., crop images for disease detection). Many rural farming areas may not have reliable or fast internet access, limiting the app's usability in such regions.

Manual Soil Nutrient Testing Without Sensors: Without the integration of soil sensors, farmers must rely on manual soil testing to obtain nitrogen, phosphorus, and potassium (NPK) levels. This can be time-consuming, costly, and may not provide real-time data, reducing the effectiveness of the Soil-Based Crop Recommendation feature.

Accuracy of Disease Detection: The disease detection system is dependent on the quality of the crop images uploaded by farmers. Factors like poor lighting, low-resolution images, or incomplete pictures of affected areas can lead to inaccurate diagnoses. This may result in the misidentification of diseases or incorrect recommendations.

Data Availability for Machine Learning: Machine learning models require large and accurate datasets to make precise predictions. The lack of comprehensive datasets for soil, climate, and crop diseases in certain regions may result in less accurate predictions and recommendations, affecting the system's overall effectiveness.

Blockchain Integration Cost: The implementation of blockchain for crop transactions between farmers and buyers involves both development and maintenance costs, which could be high. This may not be feasible for small-scale farmers or regions with limited resources, potentially hindering adoption.

User Adoption and Usability: The success of the app relies on farmers being familiar with smartphone technology and willing to use it regularly. Farmers who are less tech-savvy or have limited access to smartphones may find it difficult to adopt and use the app effectively. These limitations highlight areas that require further development, including offline functionality, enhanced sensor integration, and improved accessibility for farmers in regions with limited technology access.

1.5 Methodology of Problem solving

We propose a combined system where machine learning models are integrated with a blockchain-based marketplace. The machine learning component will consist of algorithms for: Crop Prediction: Using Random Forest and SVMs to recommend suitable crops based on soil and environmental data. Disease Detection: Employing deep learning for real-time identification of crop diseases from leaf images. Yield Forecasting: Predicting yields and providing fertilizer recommendations based on historical and real-time data. The blockchain marketplace, FarMarketplace, will support secure, transparent trade between farmers and buyers, using smart contracts to facilitate transactions and ensure fair pricing. Integration between machine learning outputs (e.g., yield forecasts) and marketplace transactions will enable more informed and efficient trading decisions.

CHAPTER 2

LITERATURE SURVEY

2.1 Literature survey

Kamilaris, A., Prenafeta-Boldú, F. X. (2018) "Machine Learning in Agriculture: A Review" Kamilaris and Prenafeta-Boldú provide a comprehensive review of the applications of machine learning in agriculture. Their survey covers a wide range of machine learning techniques that have been applied to optimize agricultural processes, from crop yield prediction to disease detection and smart irrigation systems. The study emphasizes the potential of machine learning to improve decision-making and resource efficiency in agriculture. It also highlights challenges such as the need for large datasets, data variability across regions, and the complexity of integrating various machine learning techniques into existing agricultural systems.

Mohanty, S. P., Hughes, D. P., Salathé, M. (2016) "Using Deep Learning for Image-Based Plant Disease Detection" Mohanty et al. applied deep learning techniques, particularly convolutional neural networks (CNNs), to automate the detection of plant diseases through image analysis. The study utilized a large dataset of labeled images to train the model, which outperformed traditional image processing techniques in terms of accuracy and speed. The paper emphasizes the advantages of deep learning in identifying plant diseases early, enabling timely interventions and reducing crop losses. The research also points out the scalability of deep learning models, which can be applied to various crops and disease types without significant modifications.

Wolfert, S., Ge, L., Verdouw, C., Bogaardt, M. J. (2017) "Big Data in Smart Farming – A Review" This paper explores the role of big data in smart farming and how data-driven approaches can transform agricultural practices. Wolfert et al. discuss the use of big data analytics to improve farm management, including precision farming, crop monitoring, and resource optimization. The paper highlights the potential of big data to address global food security challenges by enabling more accurate predictions and better decision-making. However, the study also points out challenges such as data privacy, the need for real-time data processing, and the integration of data from various sources.

Tian, F. (2016) "An Agri-Food Supply Chain Traceability System for China Based on RFID Blockchain Technology" Tian's study focuses on the use of blockchain and RFID technologies to create a traceability system for the agricultural supply chain in China. The research demonstrates how blockchain ensures transparency and security in tracking agricultural products from farm to consumer. By integrating RFID for real-time data collection, the system helps improve food safety, reduce fraud, and enhance

supply chain efficiency. The paper also discusses the scalability and practicality of implementing such technologies across diverse agricultural sectors, underscoring blockchain's potential to address traceability challenges globally.

Jeong, J. H., Resop, J. P., Mueller, N. D., et al. (2016) "Random Forests for Global and Regional Crop Yield Predictions" Jeong et al. utilized random forest machine learning models to predict crop yields at both global and regional levels. The study highlights the effectiveness of random forests in processing large and complex datasets, incorporating multiple variables such as climate conditions, soil properties, and historical crop data. The results demonstrate that random forests can provide accurate yield predictions, helping farmers and policymakers make informed decisions. The research also emphasizes the adaptability of machine learning models to different geographical regions, making them a valuable tool in addressing food security challenges.

Mahmood, M. R., Matin, M. A., Goudos, S. K. (2023) "Machine Learning for Smart Agriculture: A Comprehensive Survey" Mahmood et al. provide an extensive review of machine learning applications in smart agriculture, covering areas such as soil management, crop prediction, irrigation optimization, and pest control. The paper outlines various machine learning models (e.g., decision trees, random forests, SVM, neural networks) and their use cases in improving agricultural productivity. The authors also explore emerging trends, including the integration of IoT and AI, to create more efficient and sustainable farming systems. The survey highlights the importance of data-driven solutions in addressing global agricultural challenges while discussing the limitations and future prospects of machine learning in this domain.

CHAPTER 3
SOFTWARE REQUIREMENTS
SPECIFICATION

3.1 Assumptions and Dependencies

Assumptions :

Availability of Accurate Agricultural Data: The project assumes the availability of a well-curated and accurate dataset containing agricultural data, including soil characteristics (NPK levels, pH), weather conditions, and crop yield statistics. This data is essential for training machine learning models used for crop recommendation, disease detection, and yield prediction.

Diverse Soil and Crop Data: It is assumed that the collected data will be diverse, covering a wide range of soil types, crops, and environmental conditions. This diversity is crucial to ensure that the machine learning models can generalize well across different regions and farming scenarios.

Stable Internet Connectivity in Rural Areas: The project assumes that farmers using the app will have access to a stable internet connection. This is necessary for fetching real-time weather updates, accessing blockchain verification systems, and uploading crop images for disease detection.

Access to Blockchain and Weather APIs: The project presupposes the availability and seamless integration of APIs such as the Google Weather API for real-time weather data and the Marmer Marketplace blockchain platform for verifying the authenticity and traceability of agricultural products and fertilizers. These services are integral to delivering reliable and accurate information to farmers.

User Adoption of Technology: It is assumed that farmers and agricultural stakeholders will adopt and actively use the technology, including smartphones and the app, to input data, interact with the Marmer Marketplace, and access recommendations.

Sufficient Computing Infrastructure: The project assumes access to sufficient computing resources, including cloud storage and processing power, to handle large datasets and perform real-time analysis using machine learning models.

Dependencies :

Access to High-Quality Agricultural Data: The project depends on the availability of high-quality, well-labeled agricultural data, including soil nutrient levels, crop disease images, and environmental data. This data is necessary for training the machine learning models that drive crop recommendations and disease detection.

Reliable Cloud Computing and Storage Infrastructure: The project relies on access to cloud infrastructure that can store and process large datasets in real-time. This includes storage for agricultural data, weather informa-

tion, and blockchain transactions, as well as computational power for running machine learning algorithms.

Stable Internet for Real-Time Updates: The project is dependent on the availability of stable and reliable internet connections in rural areas, enabling farmers to access real-time weather forecasts, upload images for disease detection, and use blockchain-based authentication services.

Integration of Blockchain and Weather APIs: The success of the project depends on the seamless integration of external services such as the Marmer Marketplace blockchain platform for agricultural product and fertilizer authentication and APIs for weather forecasting. These services are critical for providing timely and accurate information to farmers.

User Accessibility and Education: The project is reliant on the willingness and ability of farmers to adopt the technology, including smartphones and internet-based services. Training and education on how to use the app effectively are essential for successful deployment.

Availability of Sufficient Funding and Resources: Adequate funding is required to maintain the computing infrastructure, develop machine learning models, and support blockchain integration. This includes the costs of data collection, system development, and cloud infrastructure.

3.2 Functional Requirements

3.2.1 Data Collection Requirements

1. Soil and Environmental Data for Crop Recommendation

To build an effective machine learning model for soil-based crop recommendation, the system requires the following data:

- **Soil nutrient data:** Nitrogen (N), Phosphorus (P), and Potassium (K) levels.
- **pH levels:** Indicates the acidity or alkalinity of the soil.
- **Soil type:** Texture classification such as clay, loam, or sandy.
- **Temperature data:** Historical temperature records of farming regions.
- **Humidity data:** Historical humidity levels affecting crop growth.
- **Rainfall data:** Precipitation statistics over time in various regions.

- **Electrical conductivity:** Reflects soil salinity, which affects plant growth.
- **Organic carbon content:** Indicator of soil fertility and microbial activity.

Data should be gathered from government agricultural databases, soil testing centers, IoT-based soil sensors, and satellite imagery. The dataset must encompass diverse agricultural regions to ensure model generalization.

2. Crop Disease Image Data for Disease Detection

For the disease detection module, the following image data is essential:

- **Healthy crop images:** Serve as baseline data to help distinguish between healthy and diseased plants.
- **Diseased crop images:** High-resolution images of infected leaves, fruits, or stems, covering multiple diseases (e.g., rust, blight, mildew).
- **Image diversity:** Captured under varied lighting, angles, and environmental conditions to improve generalization.
- **Meta-information:** Includes crop name, disease type, growth stage, and geotagged data.

These images can be sourced from agricultural research institutions, public datasets (e.g., PlantVillage), agricultural universities, or collected directly from farms via the mobile app.

3. Weather Data for Real-Time Forecasting

Weather forecasting in the system depends on:

- **Historical weather data:** Includes temperature, humidity, rainfall, wind speed, etc.
- **Real-time weather updates:** Acquired via APIs like Google Weather API or OpenWeatherMap.
- **Microclimate data:** Collected through in-field sensors for localized predictions.
- **Satellite-based imagery:** Helps monitor large-scale weather patterns.

Weather data must be continuously synchronized with the system to ensure accurate, up-to-date forecasting.

4. Blockchain Data for Fertilizer Verification via Marmer Marketplace

The fertilizer verification system requires:

- **Product metadata:** Information such as manufacturing date, batch number, and composition, uploaded by suppliers.
- **Blockchain transaction records:** Time-stamped records of product origin, certifications, and distribution history.
- **Supplier and distributor profiles:** Verified accounts to ensure traceability and trust in the supply chain.
- **Usage and transaction logs:** Optional data on fertilizer purchases by farmers to improve transparency.

Data sources include fertilizer manufacturers, supply chain partners, and government regulatory bodies to ensure authenticity.

5. Government Schemes and Subsidies Data

To assist farmers in accessing benefits from government programs, the following data is necessary:

- **Government policy data:** Descriptions of schemes, eligibility criteria, and application procedures.
- **Location-specific data:** Region-specific schemes tailored to the farmer's location.
- **Language and accessibility metadata:** Information in regional languages and formats for inclusivity.
- **Application status tracking:** Integration with e-governance portals for real-time updates.

Such data can be acquired from official government portals, agricultural departments, and NGOs focused on agricultural development.

3.2.2 Data Preprocessing Requirements

For soil and environmental data preprocessing, it is essential to clean the dataset by addressing missing, duplicate, or erroneous values. This involves handling gaps in NPK (Nitrogen, Phosphorus, Potassium) levels, temperature, and rainfall using techniques such as interpolation or imputation. Data

normalization or scaling ensures that variables like soil nutrients, temperature, humidity, and rainfall are on a comparable scale. Outlier detection and correction are crucial to prevent distorted predictions. Feature engineering can further enhance model performance by introducing derived variables such as soil moisture index or cumulative rainfall. Aggregating soil and environmental data into uniform time intervals (e.g., monthly averages) helps maintain consistency across the model's inputs.

For crop disease detection, image preprocessing involves resizing images to a uniform resolution (typically 224x224 pixels) while maintaining aspect ratio to prevent distortion. Background cropping ensures that only relevant portions (diseased crop areas) are used. Pixel normalization improves consistency under varying lighting conditions. Dataset augmentation techniques such as flipping, rotating, and zooming increase variability and robustness. Applying filters like Gaussian blur can reduce visual noise, while segmentation techniques (e.g., K-means clustering) can help isolate infected areas, improving classification accuracy.

In weather data preprocessing, syncing parameters like temperature, humidity, rainfall, and wind speed across time and location is critical. Missing values are handled using imputation, and smoothing techniques such as moving averages remove short-term noise. The data should be grouped into meaningful intervals (e.g., hourly or daily bins) for more actionable insights in farming decisions.

For the blockchain-based agriculture marketplace, data integrity and formatting are critical. Product listings must include standardized metadata such as crop type, quantity, pricing, location, and seller details. Transactions must be structured for compatibility with smart contracts, and any sensitive data (e.g., user credentials) should be securely encrypted. Data validation ensures that only legitimate, verified listings are recorded on the blockchain, preserving trust and transparency across the platform.

Regarding government schemes and subsidy information, preprocessing involves cleaning and organizing the data into structured formats that highlight eligibility, documentation, benefits, deadlines, and application procedures. Filtering based on the farmer's location and landholding size ensures relevance. Outdated or redundant entries should be removed to maintain current and accurate listings.

Across all datasets, data labeling is essential for supervised learning tasks, allowing models to learn meaningful patterns. Splitting datasets into training, validation, and testing sets is key to evaluating performance and avoiding overfitting. Additionally, class balancing—especially in cases like crop disease classification—ensures that the model is not biased toward a particular class (e.g., healthy vs. diseased crops).

In conclusion, effective preprocessing across all data types—soil, image, weather, blockchain, and government schemes—is foundational to the accuracy and scalability of the Smart Agriculture System. By ensuring clean, consistent, and enriched datasets, the system will deliver precise recommendations and decisions, empowering farmers to optimize crop yields, plan effectively, and engage confidently in digital marketplaces.

3.2.3 Model Requirements

The model requirements for the agricultural app include an image processing system using OpenCV to detect crop diseases through segmentation and classification of uploaded images. A machine learning model will analyze soil data (NPK, pH) to recommend suitable crops, thereby optimizing productivity. Integration with the Google Weather API will provide real-time, location-based weather forecasts, enabling farmers to plan agricultural activities effectively. A data retrieval system will deliver relevant government schemes and subsidies based on the farmer's location. Additionally, a blockchain-based organic agriculture marketplace will enable secure, transparent transactions between farmers and buyers, promoting fair trade and traceability of produce. The app will also feature secure user authentication, scalability to support a large user base, and an intuitive interface to ensure ease of use across varying levels of digital literacy.

3.3 External Interface Requirements

3.3.1 User Interfaces

The proposed agricultural app will feature a user-friendly mobile interface, enabling farmers to seamlessly interact with its various functionalities. The interface will allow users to upload crop images for disease detection, input soil data for crop recommendations, and access real-time weather forecasts. Farmers can also explore relevant government schemes and participate in the blockchain-based agriculture marketplace, where they can list their organic produce or connect directly with buyers. The interface will facilitate easy data input and retrieval, offering a simple, intuitive experience with customizable options such as soil nutrient levels and preferred crops. Additionally, the app will provide quick access to analysis results, recommendations, and marketplace insights, enhancing usability and empowering farmers to make informed, data-driven decisions efficiently.

3.3.2 Hardware Interfaces

The agricultural app will primarily run on smartphones or tablets, leveraging touchscreen input for seamless navigation and data entry. Camera functionality is crucial for capturing crop images used in disease detection, while GPS support will facilitate location-based services such as weather forecasting and access to region-specific government schemes. The app will require a stable internet connection—via Wi-Fi or mobile data—to fetch real-time weather updates and interact with blockchain systems for fertilizer verification through the Marmer Marketplace. Adequate device storage (internal or expandable) will be necessary to locally store user data, including soil parameters and analytical results. Although designed for mobile use, the app must also be compatible with cloud infrastructure or external servers to manage larger datasets and perform compute-intensive operations like machine learning-driven disease identification and crop recommendations.

3.3.3 Software Interfaces

The agricultural app will require seamless integration with several software components to ensure efficient operation. OpenCV will be used for image processing in the Crop Disease Detection module, while machine learning frameworks such as TensorFlow or PyTorch will power the soil analysis and crop recommendation models. The app will integrate external APIs, such as the Google Weather API, to provide real-time weather data for informed

decision-making. For the blockchain-based organic agriculture marketplace, blockchain libraries and smart contract tools will enable secure, transparent transactions between farmers and buyers. Data handling will be managed using libraries like NumPy and Pandas to process and analyze soil, crop, and user interaction data efficiently. SQLite or Firebase will be used for storing user profiles, farm data, and transaction logs, ensuring smooth data management and synchronization across features. The app will be optimized for both Android and iOS platforms, maximizing accessibility for farmers in diverse regions.

3.4 Nonfunctional Requirements

3.4.1 Performance Requirements

The computation requirement is mainly focused around putting a limit on total training time required for a model to train and to handle huge numpy files containing 1000s of audio file arrays. Specifically for the use case of deep learning on audio data, the following compute requirements must be satisfied:

1. **RAM Size:** Preferred 36 Gigabytes, Minimum 10 Gigabytes
The RAM plays a key role in handling numpy files at runtime. A good RAM size is essential to load two or more numpy files at once for the model to train on. The RAM also dictates the bandwidth of the data relay amount between GPU and CPU. Lower RAM sizes can lead to higher thrashing, thereby increasing the training time.
2. **RAM Generation:** Minimum DDR4
3. **GPU Memory (VRAM):** Minimum 10 Gigabytes, GDDR5X @ 11 GBPS
A larger VRAM is a must to hold a higher number of model parameters (in millions). In deep learning, training is often performed using mini-batches of data. Larger batch sizes can lead to faster convergence and better utilization of computational resources. However, larger batch sizes also require more memory.
4. **CUDA cores:** Minimum 3000
5. **CPU:** Minimum Intel i5 10th Gen @2 GHz
The CPU plays the role of coordinating all the tasks scheduled on the GPU and all the system-level operations required for it. A faster CPU is desirable to avoid any bottlenecks in data loading, preprocessing, and transfer.
6. **Storage:** Minimum 512
SSD storage is suitable for this application.
7. **Internet Speeds:** Minimum 10 Mbps
Data downloads to and from Google Drives involve transferring data sizes in the neighborhoods of 20-30 terabytes.
8. **Softwares:** The latest libraries for CUDA, Cudnn, PyTorch, and TensorFlow are favorable.

3.4.2 Safety Requirements

Ensuring the safety of the agricultural app is crucial for providing reliable and accurate results. For the Crop Disease Detection feature, the system must minimize false positives and ensure that diagnoses are accurate to avoid mis-informed treatments that could damage crops. Similarly, for the Soil-Based Crop Recommendation feature, it is essential to analyze soil data precisely to offer suitable crop suggestions, reducing the risk of crop failure. The Weather Forecasting feature must deliver real-time and dependable forecasts to support informed planning and prevent weather-related agricultural losses. In the case of the Blockchain-Based Agriculture Marketplace, it is vital to guarantee secure and transparent transactions, ensuring farmers are protected from fraud and receive fair compensation for their produce. Maintaining the integrity, precision, and trustworthiness of all data and functionalities is key to the app's safety, effectiveness, and real-world impact on farming communities.

3.5 System Requirements

The following devices were used for training the models, provided by PICT Labs. All necessary software installations and GPU enablement were performed from scratch.

1. Workstation 1

- Operating System: Ubuntu 22.04
- System RAM: 16 GB
- Storage: 512 GB SSD
- GPU: Nvidia GTX 1080 Ti (8 GB VRAM)
- Nvidia Driver Version: 535
- CUDA Version: 11.8
- cuDNN Version: 8.9.6
- TensorFlow Version: 3.13.0
- Remote Access: Enabled via AnyDesk

2. Workstation 2

- Operating System: Ubuntu 22.04
- System RAM: 16 GB

- Storage: 1 TB HDD
- GPU: Nvidia Titan X (12 GB VRAM)
- Nvidia Driver Version: 545
- CUDA Version: 11.8
- cuDNN Version: 8.9.6
- TensorFlow Version: 3.15.0
- Remote Access: Enabled via AnyDesk

Based on the above specifications, the system requirements have been categorized as follows:

3.5.1 Database Requirements

A minimum of 512 GB SSD storage is recommended to ensure fast data access and enhance system performance. If local storage is insufficient, cloud-based storage solutions such as Amazon S3 or Google Cloud Storage should be integrated. The system should also support soft storage solutions like Network-Attached Storage (NAS) or virtual storage environments for scalability and flexibility. Additionally, compatibility with Linux file readers is essential for seamless data manipulation and system integration.

3.5.2 Software Requirements

The system runs on Ubuntu 22.04 as the base operating system. For deep learning tasks, Nvidia GPUs with CUDA and cuDNN support are required. TensorFlow serves as the core deep learning framework, with version 3.13.0 used on Workstation 1 and version 3.15.0 on Workstation 2. Remote access through tools like AnyDesk is recommended for system administration and remote experimentation.

3.5.3 Hardware Requirements

The system must include:

- A minimum of 16 GB of system RAM.
- An Nvidia GPU with CUDA support (e.g., GTX 1080 Ti or Titan X).
- A minimum of 512 GB SSD storage or equivalent high-speed HDD.
- Ubuntu 22.04 or a compatible Linux-based OS.

These requirements ensure efficient execution of deep learning models and smooth development workflows.

CHAPTER 4

SYSTEM DESIGN

4.1 Model Architecture

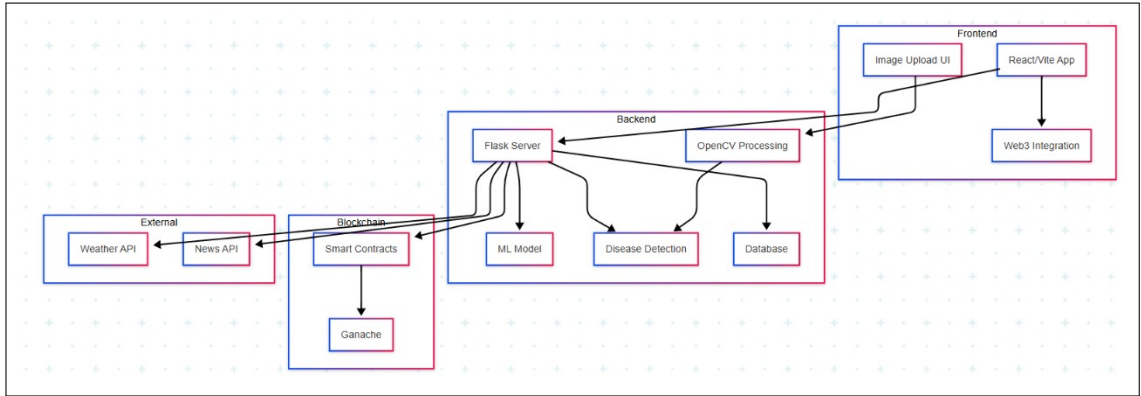


Figure 4.1: Model Architecture

The architecture of this project is designed to seamlessly integrate multiple components, including external APIs, blockchain technology, backend processing, and a modern frontend interface. It effectively combines real-time data fetching, machine learning, and decentralized smart contracts to provide a comprehensive solution. The External layer consists of the Weather API and News API, which supply real-time environmental and informational data, enriching the user experience and enhancing decision-making. These APIs feed data into the backend, ensuring the system remains responsive and context-aware.

At the core of the Backend layer is a Flask Server, which acts as the primary communication hub, coordinating various critical functions. It interfaces with a ML Model for intelligent predictions, a Disease Detection module leveraging OpenCV for image processing, and a Database for secure data storage. The backend also includes smart contracts, which connect to Ganache for blockchain simulations, ensuring secure, decentralized transactions and data integrity. This blockchain integration introduces transparency and trust to the system, crucial for handling sensitive medical data.

The Frontend layer comprises an Image Upload UI and a React/Vite App, providing a responsive and intuitive user experience. It communicates with the backend for real-time image analysis and ML processing, while the Web3 Integration allows secure interaction with the blockchain. This architecture effectively bridges traditional web technologies and modern decentralized frameworks, resulting in a powerful, flexible, and user-centric application.

4.2 UML Diagrams

4.2.1 Usecase Diagram

The use case diagram for the Agricultural Platform illustrates the interactions between different stakeholders, including Farmers, Buyers, and the System itself. Farmers can Register Crops, Track Crop growth, View Weather Forecast, and Get Crop Recommendations to improve their farming decisions. The Register Crop and Track Crop use cases are directly linked to weather data for accurate crop monitoring, leveraging external weather APIs. The Get Crop Recommendations use case also relies on real-time weather inputs to provide context-aware guidance, enhancing agricultural productivity.

Buyers primarily interact with the platform through features like List Products, Purchase Products, and Read Agriculture News, keeping them informed about market conditions and product availability. The Detect Disease use case is included to identify potential crop health issues, integrating machine learning and image processing for early detection. The System manages these processes, ensuring data consistency and security while facilitating seamless interactions among the various user roles.

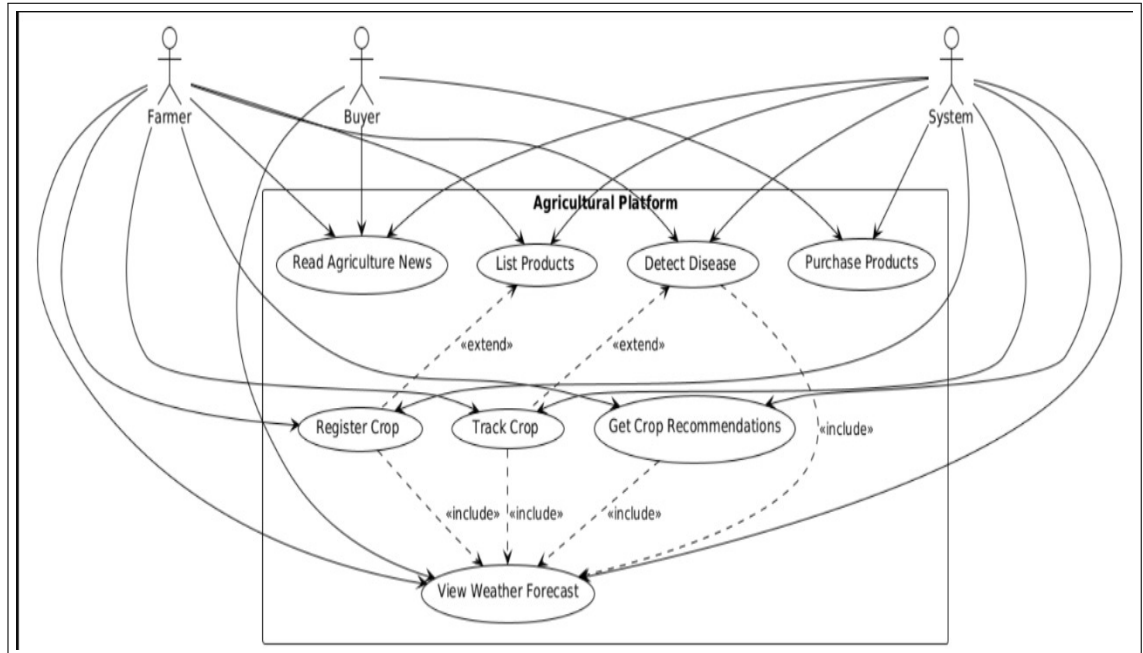


Figure 4.2: Use case diagram

4.3 Class Diagram

Class diagrams are the blueprints of the system/subsystem. Figure 4.3 models the objects that make up the system, displays the relationships between the objects, and describes what those objects do and the services that they provide.

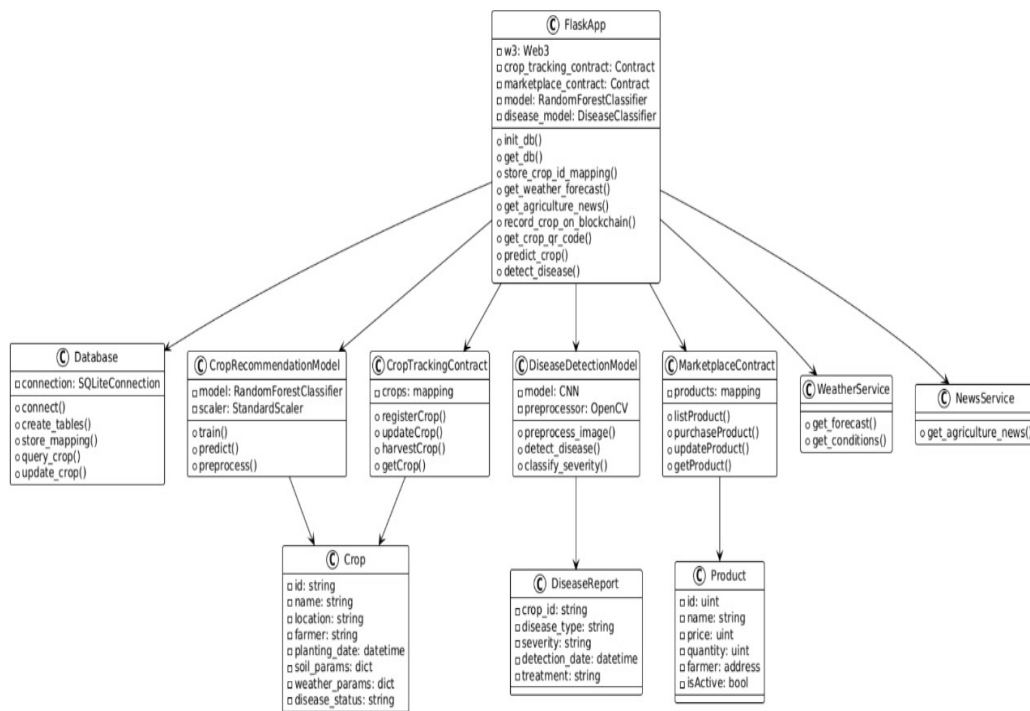


Figure 4.3: Class Diagram

CHAPTER 5

PROJECT PLAN

5.1 Risk Management

5.1.1 Risk Identification

- **Technical Risk:**
 1. Implementing accurate crop disease detection using image processing (OpenCV) may face challenges due to the diversity of crop types and disease symptoms, leading to delays or suboptimal accuracy.
 2. Integrating multiple features like weather forecasting, soil analysis, and blockchain-based verification into a single platform could lead to potential compatibility issues.
 3. The blockchain-based agriculture marketplace may face challenges in rural areas with limited internet connectivity, potentially affecting its accessibility and seamless usage for farmers.
 4. Ensuring accurate real-time weather updates in remote regions might face challenges due to inconsistent access to weather data APIs.
- **Financial Risk:** The project may experience budget overruns due to the costs associated with API integrations (e.g., Google Weather API) and the implementation of blockchain technology for secure marketplace verification.
- **Operational Risk:** Unforeseen changes in agricultural regulations or government policies could affect the app's functionality, particularly in terms of providing up-to-date information on government schemes and subsidies.
- **User Adoption Risk:** There is a risk that farmers may be reluctant to adopt the app due to a lack of familiarity with technology, especially in rural areas with limited access to digital literacy programs.

5.1.2 Risk Analysis

- **Technical Risk:** This risk is assessed as high impact, with a medium likelihood of occurrence, as technical challenges could delay project delivery. To mitigate this, thorough testing will be conducted on various crops and soil types, and pilot studies will be carried out in targeted regions to ensure real-world effectiveness.

- **Financial Risk:** The financial risk is evaluated as medium impact, with a low likelihood of occurrence. To mitigate this, cost-control measures will be enforced, and alternative, more affordable API options or open-source technologies will be explored for weather forecasting and blockchain integration in the marketplace.
- **Operational Risk:** This risk is assessed as medium impact, with a medium likelihood of occurrence. To mitigate this, a robust policy monitoring system will be implemented to ensure the app remains compliant with evolving government regulations, minimizing potential disruptions.
- **User Adoption Risk:** This risk is evaluated as high impact with a medium likelihood. To mitigate it, user-friendly design and multi-lingual, voice-activated interfaces will be developed, along with targeted digital literacy workshops to encourage adoption in rural communities.

5.1.3 Overview of Risk Mitigation, Monitoring, and Management

- **Financial Risk Mitigation:** A robust budget tracking system will be implemented to monitor expenditures on API integrations, cloud storage, and blockchain infrastructure. Cost control measures, including the evaluation and adoption of open-source alternatives to commercial APIs, will be employed to ensure the project remains within budget.
- **Operational Risk Mitigation:** A clearly defined project scope will be maintained throughout the development process. A formal change management procedure will be established to evaluate and address adjustments, especially those prompted by updates to government policies or new regulatory requirements.
- **Risk Monitoring:** Weekly risk assessment meetings will be conducted to evaluate the status of ongoing risks, including technical hurdles and financial concerns. These meetings will also serve to identify and address any emerging risks during the development and deployment phases.
- **Risk Management:** A contingency fund amounting to 10 % of the total budget will be allocated to manage unforeseen issues, such as increased API costs or delays in blockchain integration. This buffer will help ensure that the project remains on track despite unexpected challenges.

5.2 Project Schedule

5.2.1 Project Task Set

- Set up the required development environments and tools, including libraries for OpenCV, blockchain frameworks, and APIs for weather data access.
- Conduct an initial survey of publicly available datasets for crop disease images and soil nutrient data relevant to the project scope.
- Research suitable machine learning models for crop disease detection, soil analysis, and weather forecasting. Evaluate their feasibility and compatibility with the app's requirements.
- Test the accuracy and performance of the shortlisted models using curated datasets to ensure they meet the desired standards for disease detection and crop recommendation.
- Finalize the data preprocessing, model training, and evaluation workflows. Identify potential challenges or limitations and outline strategies to overcome them.
- Configure cloud-based resources to support computational demands, including remote access and environment setup for model training and data storage.
- Organize and preprocess datasets for training, ensuring correct formatting and normalization of both image and soil data.
- Train selected models on the prepared datasets, focusing on key features such as crop disease identification and soil-based crop recommendation.
- Apply necessary preprocessing techniques, including image enhancement and nutrient normalization, to improve model input quality and performance.
- Analyze training and testing results to validate model accuracy, robustness, and suitability for real-world agricultural scenarios.
- Refine and optimize model performance through iterative experimentation, adjusting parameters based on evaluation outcomes.

- Design and develop a user-friendly mobile application interface incorporating features like disease detection, soil recommendations, weather updates, and information on government schemes.
- Maintain comprehensive documentation of all project phases, including methodology, experiments, results, challenges encountered, and final outcomes.
- Conduct field/user testing with farmers to gather feedback on app usability and functionality, and implement improvements based on their suggestions.
- Prepare the project findings for submission to conferences and journals, and explore additional dissemination methods to share insights with stakeholders and the agricultural community.

5.3 Team Organization

5.3.1 Team structure

- **Kaushal Lande:** Machine Learning Engineer
- **Swapnil Adsul:** Web Developer and Blockchain Developer
- **Aditi Date :** UI/UX Designer and Web Developer
- **Mrunal Deshmukh:** Machine Learning Engineer and Blockchain Developer
- **Prof. Deepika Kumari:** Internal mentor and guide

5.3.2 Management Reporting and Communication

- **Management:** Google Workspace was utilized for workflow management, allowing for efficient tracking of task completion and allocation. Git, GitHub, and Google Colab facilitated collaboration on code development and version control.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 Overview of Project Modules

1. **Crop Prediction Module** - This module utilizes machine learning algorithms, specifically Random Forest and Support Vector Machines (SVMs), to analyze soil and environmental data. It aims to recommend the most suitable crops for planting based on a variety of factors such as soil nutrient composition, moisture levels, and climatic conditions. By providing tailored crop recommendations, this module enhances the potential for improved agricultural productivity and sustainability.
2. **Disease Detection Module** - Implementing deep learning techniques, this module focuses on the real-time identification of crop diseases through the analysis of leaf images. Using advanced computer vision methods, it can detect symptoms of diseases early, allowing farmers to take timely action to mitigate crop damage. This proactive approach helps in preserving crop health and maximizing yield.
3. **Weather Forecasting Module** - This module integrates real-time weather data using the Google Weather API to assist farmers in planning their agricultural activities effectively. By providing localized forecasts for parameters such as temperature, rainfall, humidity, and wind, it helps reduce crop damage caused by unexpected weather events. The predictive insights from this module can be further aligned with crop cycle planning and irrigation scheduling, optimizing farm management.
4. **Agriculture News Policy Module** - This module fetches the latest agriculture-related news and government policy updates using a reliable News API. It keeps farmers informed about market trends, new schemes, subsidy announcements, and regulatory changes. By delivering curated, regional-language-friendly content, it enhances awareness and empowers farmers to make timely and informed decisions in a dynamic agri-policy environment.
5. **Blockchain Marketplace Module** - This module serves as a secure and transparent trading platform for farmers and buyers. Utilizing blockchain technology, it facilitates transactions through smart contracts, ensuring fair pricing and reducing the risk of fraud. The integration of machine learning outputs, such as yield forecasts and crop recommendations, with the marketplace enables informed trading decisions, improving market efficiency and farmer profitability.

6.2 Tools and Technologies Used

- Python 3.9
- TensorFlow
- PyTorch
- scikit-learn
- NumPy
- Pandas
- Matplotlib
- Seaborn
- OpenCV
- Flask/Django (for web development)
- Blockchain frameworks (e.g., Ethereum or Hyperledger)
- Jupyter Notebook
- Google Colab
- Git GitHub
- Google Workspace
- APIs (e.g. Weather API)

6.3 Model Selection

From the analysis of various machine learning and deep learning models applicable to agricultural predictions and disease detection, several architectures were identified as promising for implementation. These models include:

Random Forest and Support Vector Machines (SVMs) for Crop Prediction: These algorithms were chosen for their robustness in handling complex datasets, allowing the app to recommend suitable crops based on soil and environmental data.

Deep Learning Models for Disease Detection: Models such as Convolutional Neural Networks (CNNs) were selected to enable real-time identification of crop diseases from leaf images. This approach leverages image recognition capabilities to enhance diagnostic accuracy.

To enhance the overall performance, the architecture incorporates a Blockchain-based Marketplace component, termed FarMarketplace, which supports secure and transparent transactions between farmers and buyers. This integration ensures that machine learning outputs, such as yield forecasts, directly inform trading decisions, optimizing the supply chain.

This multi-model approach underscores the understanding that addressing the challenges of agriculture, such as crop disease detection and yield prediction, requires a combination of different models tailored to specific tasks within the broader context of agricultural sustainability.

CHAPTER 7

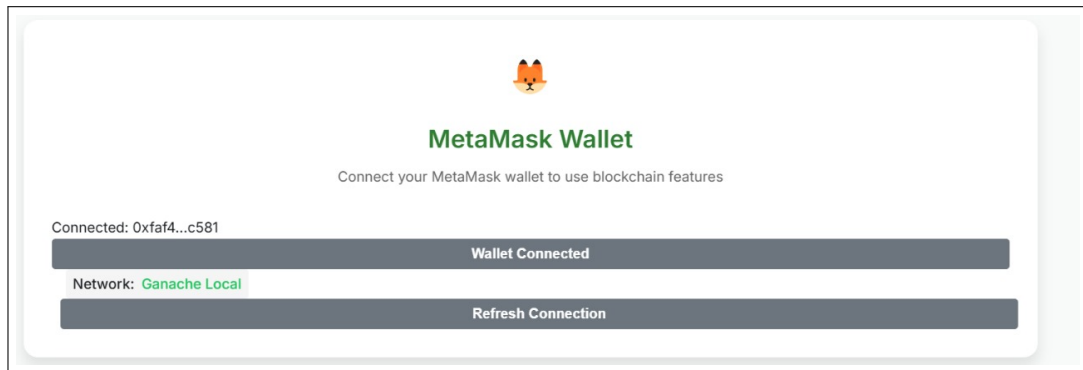
RESULTS

7.1 Outcomes

The outcomes of the project address several significant challenges that were successfully navigated:

1. **Handling Soil Variability for Crop Recommendation:** The project successfully addressed the challenge of soil variability by developing a soil-based crop recommendation system that can analyze essential nutrients like nitrogen and phosphorus. This feature provides accurate crop suggestions tailored to different soil compositions, improving overall crop productivity.
2. **Integration of Multiple Features into a Single Application:** A significant challenge was integrating multiple functionalities—crop disease detection, soil-based crop recommendations, weather forecasting, and fertilizer authenticity verification—into a single app. The project successfully implemented a unified solution, allowing farmers to access all these services seamlessly within one platform.
3. **Overcoming Limited Access to Blockchain Technology:** Despite the complexity and limited accessibility of blockchain technology in rural areas, the project successfully implemented a blockchain-based organic agriculture marketplace. This enables farmers to directly connect with buyers through a transparent and secure platform, fostering trust, improving market access, and ensuring fair pricing for organic produce.
4. **Limited Awareness of Government Schemes:** The project effectively addressed the challenge of limited awareness by providing farmers with easy access to up-to-date information on relevant government schemes and subsidies. This feature ensures that farmers can take full advantage of the available support to improve their agricultural practices.
5. **Working with Limited Connectivity in Rural Areas:** To ensure the app's usability in areas with limited internet connectivity, the project optimized the app's performance for low-bandwidth environments. This guarantees that farmers in remote regions can still access vital features like weather updates and crop recommendations even with intermittent network access.

7.2 Screen Shots



The screenshot shows the "Crop Prediction" form. At the top, there is a plant icon and the text "Crop Prediction". Below this, it says "Phosphorus content in mg/kg and environmental conditions below". The form contains several input fields for various parameters:

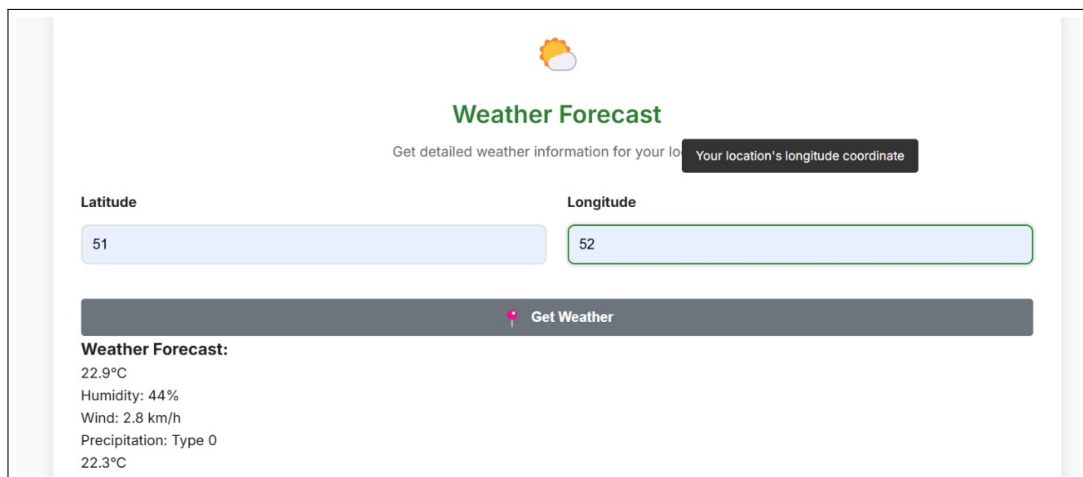
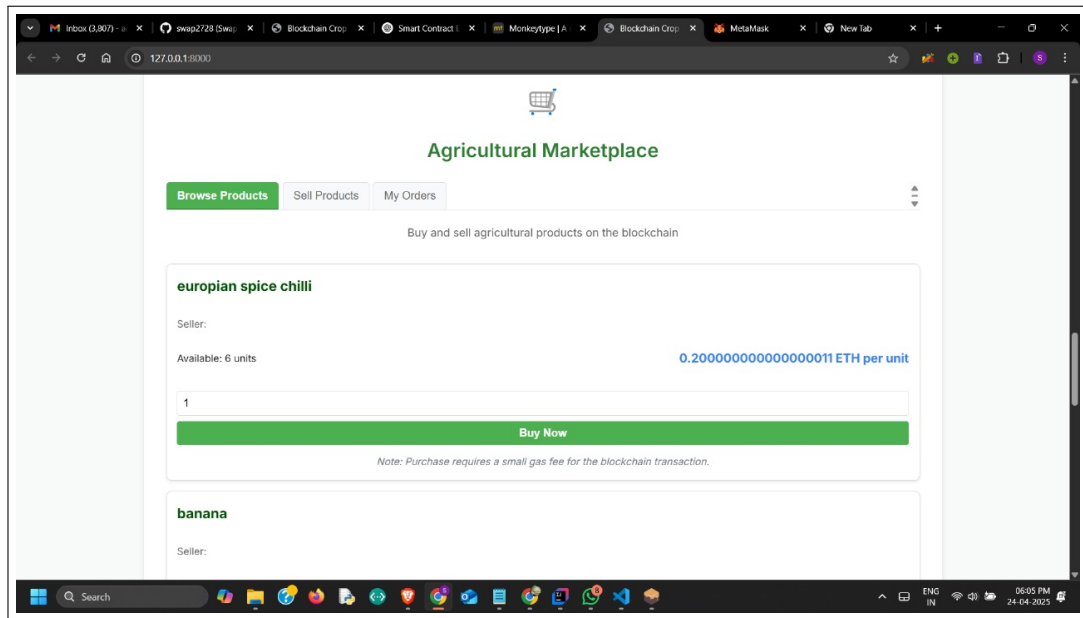
- Nitrogen (N): e.g., 40
- Phosphorus (P): e.g., 50 (with a tooltip "Please fill out this field.")
- Potassium (K): e.g., 30
- Temperature (°C): e.g., 25
- Humidity (%): e.g., 70
- pH: e.g., 6.5
- Rainfall (mm): e.g., 200

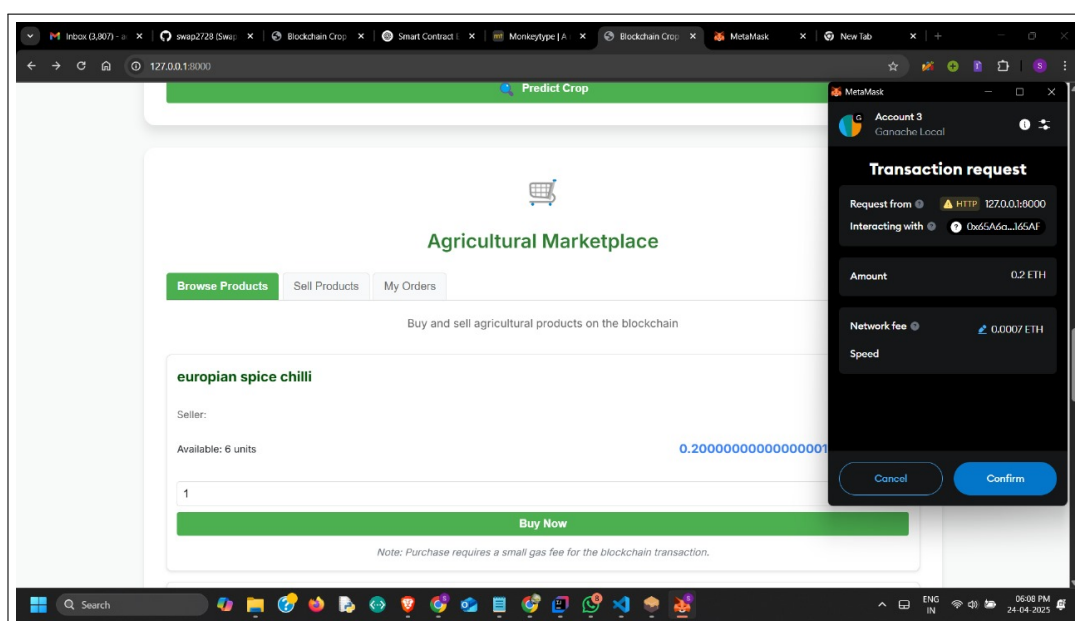
Below these fields, there is a "Blockchain Record (Optional)" section with two input fields:

- Farm Location: e.g., Maharashtra, India
- Farmer Name: e.g., John Doe

At the bottom, there is a green button labeled "Predict Crop".

Smart Agriculture System Using Machine Learning, Blockchain





CHAPTER 8

CONCLUSIONS

8.1 Conclusions

In conclusion, our integrated web application leverages advanced technologies such as computer vision, machine learning, real-time data APIs, and blockchain to address critical challenges faced by farmers. While many agricultural tools focus on isolated solutions, our approach provides a unified platform that supports informed decision-making across crop health, soil suitability, weather forecasting, and market access. Notably, most digital agricultural solutions overlook small-scale and organic farmers, particularly in resource-constrained settings. Our blockchain-based organic farming marketplace directly addresses this gap by enabling transparent, secure, and traceable transactions between farmers and buyers. Furthermore, the application is designed with low computational requirements, making it accessible and scalable even in rural or low-connectivity regions. By empowering farmers with tailored recommendations and trusted digital tools, our project not only enhances productivity and sustainability but also promotes inclusivity in the agri-tech landscape.

8.2 Future Work

1. **Integrating Advanced Soil Analysis:** Current soil-based crop recommendations rely on basic nutrient analysis like nitrogen and phosphorus. Future work could involve integrating more advanced parameters, such as soil microbial activity, organic matter content, and micronutrients, offering farmers more accurate and holistic crop recommendations.
2. **Blockchain Scalability and Cost Efficiency:** While blockchain is used for marketplace transactions, future work could explore optimizing blockchain protocols to reduce computational costs and energy consumption. This would make the solution more scalable and accessible to small-scale farmers who may have limited resources.
3. **Incorporating Real-Time Weather Alerts:** Though the app currently uses the Weather API for forecasts, incorporating real-time weather alerts based on localized data (e.g., IoT sensors) would improve the accuracy and timeliness of weather-related notifications, helping farmers take preventive action more efficiently.
4. **Improving User Accessibility:** Future iterations of the app could focus on creating a multi-lingual, voice-activated interface to cater to

farmers with limited literacy or technical skills. This feature would make the app more user-friendly, especially in rural and remote areas.

5. **Real-Time Disease Diagnosis:** Currently, the disease detection system is based on image uploads. Future developments could involve real-time disease diagnosis using video input or integrating IoT devices equipped with cameras in the fields, allowing for continuous monitoring and immediate alerts when diseases are detected.
6. **Data Privacy and Security Enhancements:** As the app collects sensitive data such as soil composition and crop health, future work should prioritize data privacy and security by enhancing encryption methods. This ensures farmers' data is protected, while also maintaining the transparency and trust enabled by blockchain.

8.3 Applications

The noise suppression model can be applied in various contexts:

1. **Soil-Based Crop Recommendation:** Selecting the right crops based on soil composition is crucial for maximizing yield. The app analyzes soil nutrient data, such as nitrogen and phosphorus levels, to recommend the most suitable crops for farmers, optimizing both productivity and sustainability.
2. **Weather-Based Activity Planning:** Weather plays a critical role in farming activities. By integrating real-time weather forecasts from the Google Weather API, the app assists farmers in planning their activities—such as irrigation, planting, and harvesting—more effectively, reducing the risk of crop loss due to unfavorable weather conditions.
3. **Blockchain-Based Organic Agriculture Marketplace:** Farmers often face challenges in reaching trustworthy buyers and getting fair prices for their produce. The app's decentralized marketplace, built using blockchain technology, connects organic farmers directly with consumers and retailers. It ensures transparency, traceability, and secure transactions by maintaining immutable records of farming practices and product certifications.
4. **Access to Government Schemes:** Many farmers are unaware of the government schemes and subsidies available to them. By providing access to detailed information on relevant policies and schemes, the app

helps farmers take full advantage of resources designed to support their agricultural activities.

CHAPTER 9

APPENDIX

9.1 Appendix A

9.1.1 Technical Feasibility

- The project leverages advanced frameworks such as OpenCV for image processing and TensorFlow for model development, ensuring technical feasibility in implementing crop disease detection based on image inputs.
- Integration of external APIs, like the Google Weather API, for real-time weather forecasting demonstrates the technical viability of incorporating third-party services into the app.
- The use of blockchain technology for the organic agriculture marketplace ensures a secure, transparent, and scalable solution, further supporting the technical feasibility of the project.
- The crop recommendation system is built on soil nutrient analysis models, employing lightweight algorithms to efficiently process data, making the system technically feasible even in resource-constrained environments.

9.1.2 Operational Feasibility

- Preliminary user testing and feedback from farmers and agricultural experts indicate that the project is operationally feasible in real-world farming scenarios.
- The seamless integration of soil data collection, weather forecasting, and crop disease detection workflows optimizes operational processes, ensuring that farmers can use the app efficiently.
- The development of a simple, intuitive user interface allows for easy navigation and access to critical information, enhancing operational feasibility, especially for users with limited technical expertise.

9.1.3 Economic Feasibility

- A cost-benefit analysis was conducted, factoring in expenses for app development, integration of machine learning models, blockchain technology, and third-party API services.

- The economic benefits, such as improved crop yield through early disease detection, optimized crop selection based on soil quality, and prevention of counterfeit fertilizer use, significantly outweigh the costs, confirming the project's economic feasibility.
- Additionally, the potential long-term savings for farmers, coupled with increased productivity and resource optimization, further justify the economic viability of the project.

9.2 Appendix B

9.2.1 Survey Paper Details

Paper Status: Published

9.3 Appendix C

9.3.1 Plagiarism Report

The plagiarism report for the project report has been attached herewith.

Similarity: 1%

CHAPTER 10

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
API	Application Programming Interface
DNN	Deep Neural NetworkTransform
SVM	Support Vector Machine
RF	Random Forest
ML	ML: Machine Learning
PESQ	Perceptual Evaluation of Sound Quality
GPU	Graphics Processing Unit